

How to choose a data visualization

If your data has a changing variable

You can use these visualizations

Which look like this

Line charts

Individual data points for a changing variable are connected with a continuous line

Download a [stacked line chart](#) in Google Sheets

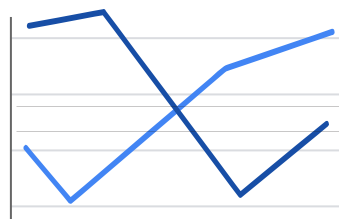
Single:

when the changing variable is for a single category



Stacked:

when the changing variable applies to more than one category and you want to compare categories



Column charts

(vertical bar charts)

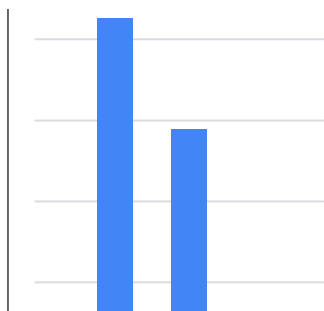
Individual data points for a changing variable are represented as vertical columns

Note: If the values being compared are vastly different, a column chart might be too tall. You can use a horizontal bar chart instead.

Download [examples](#) in Google Sheets

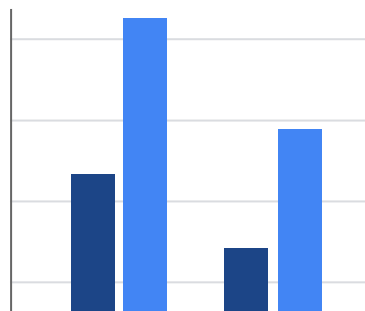
Single:

when the changing variable is for a single category



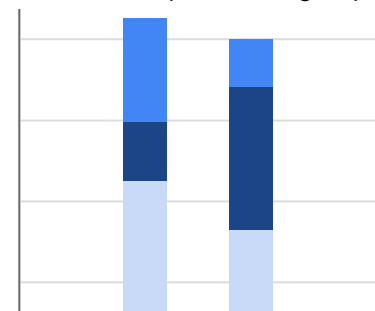
Grouped:

when the variable change applies to more than one category and you want to compare categories



Stacked:

when the variable change applies to more than one category and you want to compare categories without the spread of a group



Horizontal bar charts

Individual data points for a changing variable for one or more categories; these appear like rotated column charts

Download [examples](#) in Google Sheets

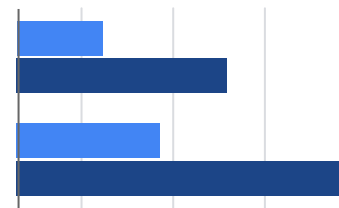
Single:

when the changing variable is for a single category



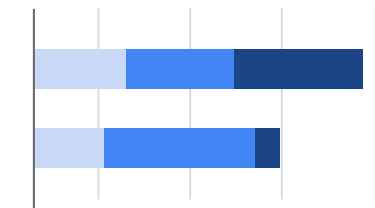
Grouped:

when the variable change applies to more than one category and you want to compare categories

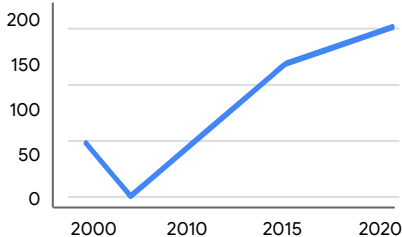
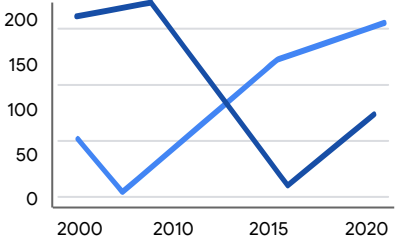
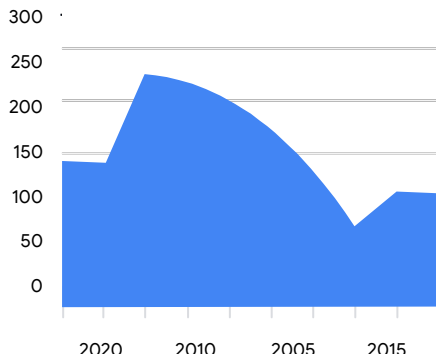
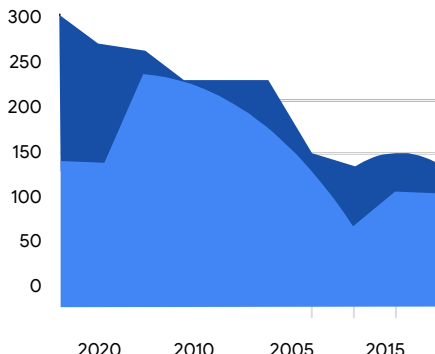
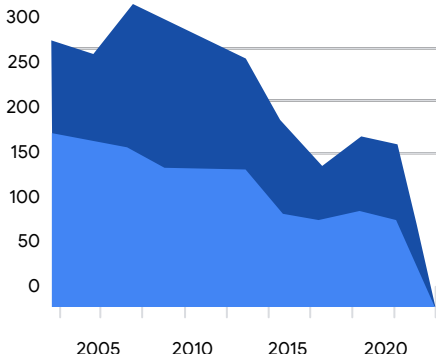


Stacked:

when the variable change applies to more than one category and you want to compare categories without the spread of a group



How to choose a data visualization

If your data has a changing variable measured over time		
You can use these visualizations	Which look like this	
Line charts Individual data points for a changing variable are connected with a continuous line Download a stacked line chart in Google Sheets	The line charts are similar to those for a changing variable but time is shown on the x-axis	Single: when the change over time is for a single item or classification  Stacked: when the change over time is for multiple items or classifications 
Area charts Individual data points for a changing variable are connected with a continuous line and the area under the line is filled in Download a stacked area chart in Google Sheets	Single: when the variable change is for a single category over time 	Unstacked: when data doesn't align on the x-axis (data is from different time points)  Stacked: when data aligns on the x-axis (data is from the same time points) 

How to choose a data visualization

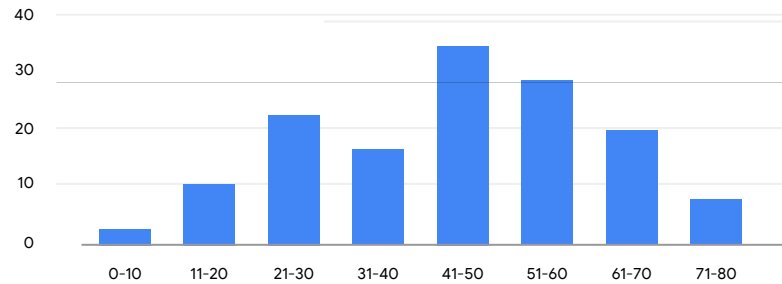
If your data has a numeric trend

You can use these visualizations

Histograms

Individual data points are categorized into columns that each represent a different range of values

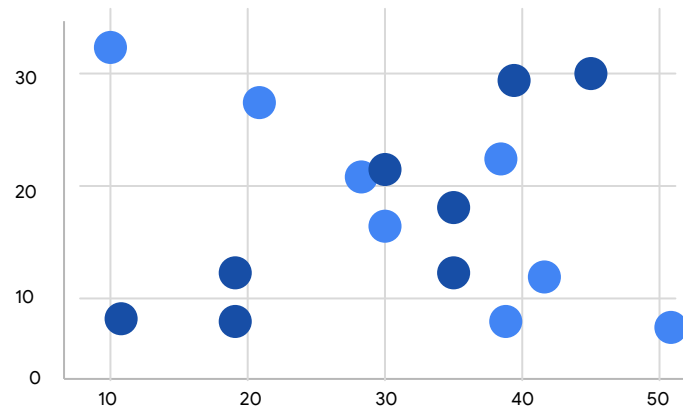
Download a [histogram](#) in Google Sheets



Scatter charts

Individual data points are displayed, but without a connecting line like in a line chart

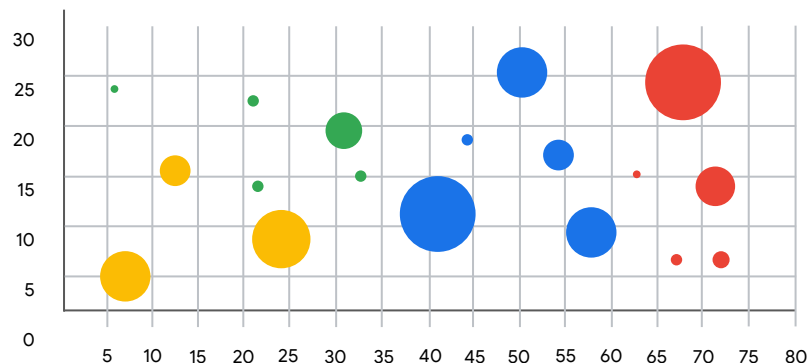
Download a [scatter chart](#) in Google Sheets



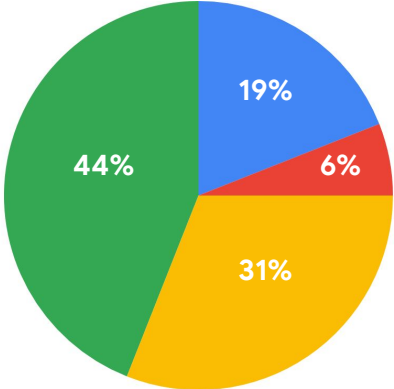
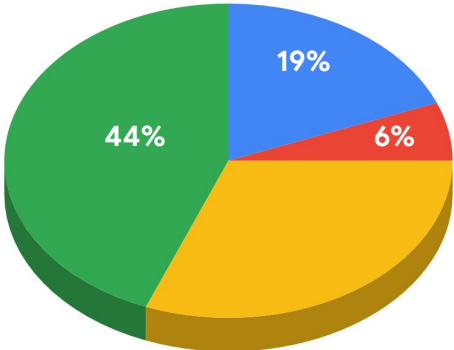
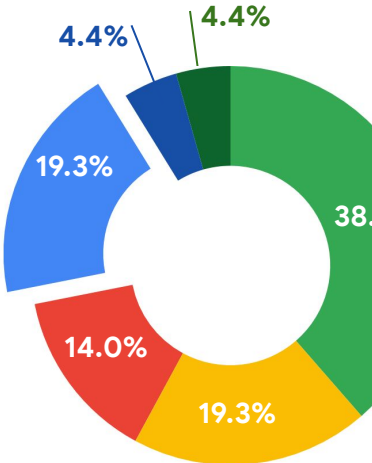
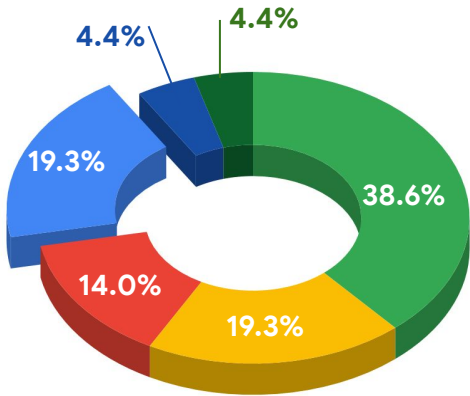
Bubble charts

Individual data points are displayed as bubbles like in a scatter plot, but numeric values are compared relative size of the bubbles

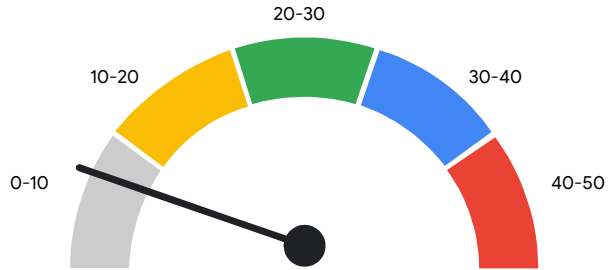
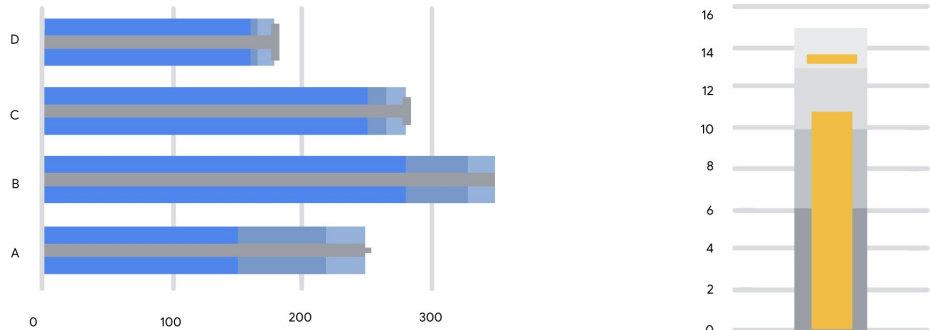
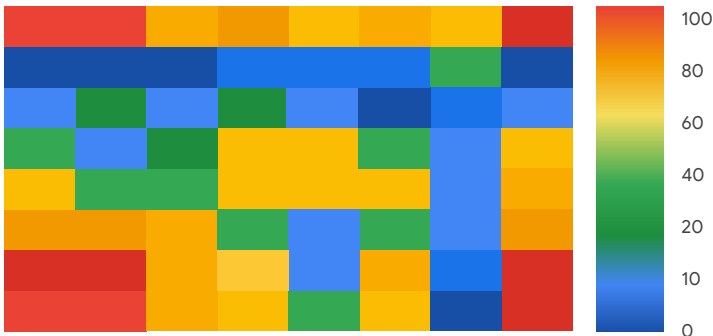
Download a [bubble chart](#) in Google Sheets



How to choose a data visualization

If your data has partial and whole results	
You can use these visualizations	Which look like this
Pie charts 2D or 3D proportions (slices) are shown adding up to a whole or 100% Download a 2D pie chart in Google Sheets	Two-dimensional:  Three-dimensional: 
Donut charts 2D or 3D proportions (segments) adding up to a whole or 100% Download a 2D donut chart in Google Sheets	Two-dimensional:  Three-dimensional: 

How to choose a data visualization

If your data is progressive	
You can use these visualizations	Which look like this
Gauge charts Single result is shown within a progressive range of values allowed Download gauge charts in Google Sheets	
Bullet charts Progressive result is shown as a horizontal or vertical bar chart moving towards a desired value	
If your data has intensity or frequency	
You can use these visualizations	Which look like this
Heat maps Results are shown by color gradations representing the strength or frequency of values; higher or more frequent values have more intense color	

How to choose a data visualization

If your data has intensity or frequency (continued)	
You can use these visualizations	Which look like this
Density maps Results are shown by color representing the number or frequency of data points in a given area on a map	